

Package ‘GenotypeQuant’

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Type Package

Title Estimates the number of genotypes present in a eDNA sample
genotyped with microsatellite markers

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Description The method uses the count of unique (different) microsatellite alleles amplified from an eDNA sample to estimate the number of genotypes present in the sample using a numerical maximum likelihood approach.
In other words, the estimate is the number of genotypes that most likely generated the observed counts of different alleles for each locus. The method relies on resampling an allele frequency table for the reference population to generate the distribution of unique allele numbers expected in simulated samples with different numbers of gametophytes.
Users can estimate the number of haploid or diploid genotypes. A correction for FIS is available.

Imports adegenet, genepop

License GPL (>=2)

NeedsCompilation no

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G.Estimate

*Estimates the most likely number of gametophytes in an eDNA sample***Description**

The function compares a user supplied vector with number of different alleles observed (genotyped) for each locus with sampling distributions created from a reference population [MakeAllNumberDist](#). A numerical maximum likelihood method is used to estimate the most likely number of haploid (or diploid) genotypes that might have generated the observed number of different alleles observed across loci.

Usage

```
G.Estimate(Obs.N.all, All.Num.Dist, ML.curve=TRUE, ploidy=1,
col="black", col.est="darkcyan", col.ci="darkcyan")
```

Arguments

Obs.N.all	A vector with length equal to the number of locus in the reference population allele frequency file. The values are the total number of different (unique) alleles genotyped at each locus from an eDNA sample. Locus with missing data should be coded with NA.
All.Num.Dist	The 3-D array produced by MakeAllNumberDist . This array contains the sample distributions with unique allele counts generated by sampling the reference population allele frequencies, with or without Fis correction.
ML.curve	A logical to decide if a plot with the maximum-likelihood curve should be produced; defaults to TRUE.
ploidy	If estimating the number of haploids genotypes set to 1, if diploids set to 2.
col	The color of the ML-curve line.
col.est	The color of the vertical line intersecting the ML-curve at the estimated number of gametophytes.
col.ci	The color of the vertical lines intersecting the ML-curve at the 95 percent confidence interval.

Details

The probability that a certain number of observed alleles $n_{obs.A}$ at locus l was produced by g gametophytes is estimated by the relative frequency of $n_{obs.A}$ in the sampling distribution for that locus and g gametophytes, recorded in **All.Num.Dist**. The maximum likelihood is found by summing the log probabilities for all loci. An approximate confidence interval is produced using the likelihood ratio approach; in brief, the 95 percent interval includes all gametophytes values for which the log-likelihood function drops off by no more than 1.92 units.

Value

A list with the following components:

estimateM	A vector with three elements, the most likely number of gametophytes (G.hat), and the lower and upper bounds of the 95 percent confidence interval.
LogML	The log-likelihoods for all values of gametophytes from one to the maximum number available in the sampling distributions. Inf values are returned when <i>n.obs.A</i> are impossible given a particular number of gametophytes.

Author(s)

F. Alberto and L. Liggan.

References

Liggan L, Alberto F (in prep)...

See Also

[MakeAllNumberDist](#)

MakeAllNumberDist	<i>Generates distributions of total number of alleles that can be produced by different number of gametophytes.</i>
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Description

By resampling from the allele frequency table for the reference population, this function generates distributions of total number of alleles that can be produced by different number of gametophytes.

Usage

```
MakeAllNumberDist(genInd,ngametos=80,npermutations=5000)
MakeAllNumberDist(genotypes.df,Loci.names,ngametos=80,All.nchar=3,Fis=FALSE,Fis.threshold=0.1,npermu
```

Arguments

genotypes.df	The input file with the reference population microsatellite allele codes. This is a data.frame with one column per locus one row per individual with column names containing the locus names and without individual information (i.e. the first column is the information for the first loci). Use the same number of digits to code each allele. For example, 120126 is a possible value for a diploid coded with three digits per allele. At this stage, the reference data needs to be in diploid format. Missng data should be coded with the number zero.
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ngametos	The method will generate distributions of total number of unique alleles that could have been generated by samples containing one to g number of haploid genotypes. Distributions are sampled from the reference allele frequency table. There are distributions for each combination of locus and number of gametophytes.
All.nchar	How many digits are used in the reference population file to code each allele (can only be 1, 2, or 3).
Fis	A logical indicating if Fis corrections should be applied when generating the sampling distributions. If yes, Genepop is used to calculate Qintra, the observed frequencies of identical pairs of genes in the reference populations. Alleles are then sampled sequential with probability Qintra to sample the previously sampled allele and 1-Qintra to sample any other allele according to their allelic frequencies. Defaults to FALSE.
Fis.threshold	When applying Fis correction is advisable to only correct for Fis values above a certain threshold. The default is 0.1.
npermutations	How many sample values should each distribution contain; defaults to 5000.

Details

The function creates distributions of total number of alleles that are likely to be produced for each loci for different numbers of gametophytes (from one to ngametos).

Value

A three-dimensional array with dimensions ngametos, number of loci (in the reference population allele frequency table), and npermutations, containing values for the total number of alleles sampled. This object will be required by the function [G.Estimate](#) to estimate the most likely number of gametophytes present in the eDNA sample.

Author(s)

F. Alberto and L. Liggan.

References

Liggan L, Alberto F (in prep)...

See Also

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